Varun A. Paturkar

PhD Student, Computer Science and Engineering, CVIT, IIIT Hyderabad

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Research Interests

Safe and reliable perception and reasoning for autonomous driving in real-world conditions, with a current focus on building and evaluating trustworthy vision-language models for driving.

Education

Ph.D., Computer Science and Engineering CGPA 8.75/10 Aug 2025 – Present
CVIT, IIIT Hyderabad. Advisor: Prof. C. V. Jawahar

B.E., Mechanical Engineering CGPA 7.37/10 Aug 2016 – Jul 2020
Sir M. Visvesvaraya Institute of Technology, Visvesvaraya Technological University, Bengaluru

Publications

[1] **Varun A. Paturkar**, Shankar Gangisetty, C. V. Jawahar. *MotoVerse: A Benchmark for Predictive Perception of Motorized Two-Wheelers in Dense Unstructured Traffic*. **WACV** 2027 (Datasets Track, to appear).

[2] **Varun A. Paturkar**, Shankar Gangisetty, C. V. Jawahar. *MOTOR: A Multimodal Dataset for Two-Wheeler Rider Behavior Understanding*. **ICRA** 2026. [Project Page] [Paper] [Dataset]

[3] **Varun A. Paturkar**, R. Yadav, R. Kala. *Sequential Place Recognition using Semantically Enhanced Images*. *Multimedia Tools and Applications*, 2023.

[4] **Varun A. Paturkar**, S. N. Murali, S. S. Nambiar, Shankar Gangisetty, C. V. Jawahar. *LadderDrive: Evaluating Granularity and Reliability of Vision Language Models in Driving Scenarios*. Under review, 2026.

Experience

PhD Researcher Aug 2025 – Present
CVIT, IIIT Hyderabad

- **LadderDrive** (with Bosch Research India): an object-centric diagnostic benchmark that tests the granularity and reliability of driving VLMs and VLAs within perception, prediction and planning; 80 scenes, 31,102 human-verified annotations, eight models evaluated.
- **MotoVerse** (WACV 2027): a benchmark for anticipating a rider's manoeuvre and forecasting the paths of surrounding traffic from a single front camera; 949 clips, 31,000+ road agents, 16 models benchmarked.

Research Intern Jan 2025 – Aug 2025
CVIT, IIIT Hyderabad

- **MOTOR** (ICRA 2026): a multimodal dataset of two-wheeler riding (multi-view video, rider eye gaze, GPS/IMU; 1,629 sequences from 16 riders) with manoeuvre and legality labels; Showed that fusing gaze and vehicle motion with video improves recognition. Dataset publicly released.

Research Assistant (Part-time) Aug 2021 – Mar 2023
IIIT Allahabad, with Prof. Rahul Kala

- Sequential visual place recognition with semantically enhanced features, outperforming several state-of-the-art methods on a self-collected dataset and Oxford RobotCar.

Machine Learning Engineer Jan 2021 – Dec 2025
Navajna Technologies, Hyderabad

- **Driver Monitoring System** (with Mercedes-Benz R&D India): defined data requirements and the sensor setup, built pre-processing and automated quality-check pipelines, and led a 15-member team across 5 countries; 800 TB collected at 98% usability.
- **Crowd safety analysis** (with Prof. Dinesh Manocha): led video analysis of crowd density, speed and direction for a music-festival crowd disaster.
- Also built a conversational avatar deployed with Intel OpenVINO (85% faster inference) and a highway-condition assessment system to NHAI guidelines.

Talks, Awards & Service

- Paper presentation, **MOTOR**, ICRA 2026, Vienna; Supported by an **IEEE RAS Travel Grant**
- Tutorial on **Transformers**, CVIT Summer School, IIIT Hyderabad, 2026
- Talk, **"From Pixels to Actions"**, CIE-IIITH CV Labs Workshop (8th Edition), 2026
- Reviewer: ICRA, IROS, WACV, ACCV | Certified Artificial Intelligence Professional, DIAT (DRDO), 2022